# Verified experimental comparisons

CLIP ViT-B/32. Accuracy (%). Prepared 22 October 2026. Each result uses its matching unprotected control. Historical tests have been observed. Missing experiments are explicitly marked; no eight-task average is reported.

Table 1: Main comparison: two experts, TA coefficient 0.5

| Protection | Standalone M | No recovery M / S | AGA target M / S | AGA both M / S |
|---|---|---|---|---|
| Unprotected | 99.69 | 99.37 / 92.50 | 99.30 / 92.78 | 99.32 / 91.77 |
| PaRaMS (full) | 99.69 | 8.92 / 9.15 | 99.30 / 92.79 | 99.32 / 91.77 |
| Defense 1 | 99.69 | 9.80 / 6.70 | 9.80 / 6.70 | 9.80 / 6.70 |
| O/B [H] | 99.71 | 78.77 / 62.94 | 79.53 / 65.82 | 77.85 / 61.42 |
| A/B [H] | 99.65 | 91.07 / 77.46 | 85.42 / 75.50 | 79.20 / 69.84 |
| Defense 2 (structure) [K] | 99.69 | 7.39 / 7.74 | 10.09 / 6.78 | 10.09 / 6.78 |
| Defense 2 candidate (failed) [F] | 99.69 | 99.13 / 91.41 | 99.01 / 91.59 | 99.04 / 90.51 |

M: protected MNIST; S: partner SVHN. AGA target/both denotes recovery of only the protected expert/both experts. [H] Historical O/B and A/B fail the original complete-task bound; effect references only. [K] Known global symmetry control, not an original method; family-aware public recovery is reported separately. [F] Latest candidate preserves the complete task check but does not provide useful protection. Full PaRaMS is the full-default baseline; parameter scales differ and are not a universal exclusion rule.

Table 2: Matched defenses across three protected tasks

| Target / partner | Protection | Standalone | No recovery T / P | AGA target T / P | AGA both T / P |
|---|---|---|---|---|---|
| MNIST / SVHN | Unprotected | 99.69 | 99.37 / 92.50 | 99.30 / 92.78 | 99.32 / 91.77 |
| MNIST / SVHN | Defense 1 | 99.69 | 9.80 / 6.70 | 9.80 / 6.70 | 9.80 / 6.70 |
| MNIST / SVHN | PaRaMS (full) | 99.69 | 8.92 / 9.15 | 99.30 / 92.79 | 99.32 / 91.77 |
| MNIST / SVHN | Defense 2 (structure) | 99.69 | 7.39 / 7.74 | 10.09 / 6.78 | 10.09 / 6.78 |
| DTD / MNIST | Unprotected | 79.41 | 69.31 / 99.45 | 68.40 / 99.45 | 68.78 / 99.42 |
| DTD / MNIST | Defense 1 | 79.41 | 2.13 / 21.09 | 2.13 / 21.09 | 2.13 / 21.05 |
| DTD / MNIST | PaRaMS (full) | 79.41 | 2.07 / 8.92 | 68.35 / 99.45 | 68.78 / 99.42 |
| DTD / MNIST | Defense 2 (structure) | 79.41 | 2.23 / 8.68 | 2.18 / 9.89 | 2.13 / 9.85 |
| SVHN / MNIST | Unprotected | 97.46 | 92.50 / 99.37 | 91.56 / 99.39 | 91.77 / 99.32 |
| SVHN / MNIST | Defense 1 | 97.46 | 6.70 / 9.80 | 6.70 / 9.80 | 6.70 / 9.80 |
| SVHN / MNIST | PaRaMS (full) | 97.46 | 9.09 / 8.95 | 91.56 / 99.39 | 91.77 / 99.32 |
| SVHN / MNIST | Defense 2 (structure) | 97.46 | 7.74 / 7.39 | 9.74 / 10.28 | 9.63 / 10.28 |

TA0.5 and unchanged original AGA; every merged cell is protected / partner accuracy. Full PaRaMS uses official defaults without clipping. Defense 2 uses one fixed data-free global permutation rule; task-SVD-guided construction and novelty remain under study. Three protected tasks correspond to two unordered task pairs. Known-family/location-aware recovery is a separate permission setting. These are matched evaluations, not a complete eight-task table; complete original-task predictions are retained. Parameter scales and numerical protocols are reported separately.

Table 3: Defense 1 across three protected tasks

| Target / partner | Protection | Standalone | No recovery T / P | AGA target T / P | AGA both T / P |
|---|---|---|---|---|---|
| MNIST / SVHN | Unprotected | 99.69 | 99.37 / 92.50 | 99.30 / 92.78 | 99.32 / 91.77 |
| MNIST / SVHN | Defense 1 | 99.69 | 9.80 / 6.70 | 9.80 / 6.70 | 9.80 / 6.70 |
| MNIST / DTD | Unprotected | 99.69 | 99.45 / 69.31 | 99.43 / 69.41 | 99.42 / 68.78 |
| MNIST / DTD | Defense 1 | 99.69 | 9.80 / 2.13 | 9.80 / 2.13 | 9.80 / 2.13 |
| DTD / MNIST | Unprotected | 79.41 | 69.31 / 99.45 | 68.40 / 99.45 | 68.78 / 99.42 |
| DTD / MNIST | Defense 1 | 79.41 | 2.13 / 21.09 | 2.13 / 21.09 | 2.13 / 21.05 |
| SVHN / MNIST | Unprotected | 97.46 | 92.50 / 99.37 | 91.56 / 99.39 | 91.77 / 99.32 |
| SVHN / MNIST | Defense 1 | 97.46 | 6.70 / 9.80 | 6.70 / 9.80 | 6.70 / 9.80 |

TA0.5. Every merged cell is target / partner accuracy. The task role is fixed by the first column, not by a global dataset order. The original AGA does not modify the final LayerNorm bias; these results do not imply resistance to location-aware or data-assisted recovery.

Table 4: Recovery permissions: protected MNIST and partner SVHN

| Recovery pipeline | Additional access | Matched control M / S | Defense 1 M / S |
|---|---|---|---|
| No recovery | Published expert and partner | 99.37 / 92.50 | 9.80 / 6.70 |
| AGA target | Public base; defense-agnostic | 99.30 / 92.78 | 9.80 / 6.70 |
| AGA both | Public base; defense-agnostic | 99.32 / 91.77 | 9.80 / 6.70 |
| Diagonal adaptation | Specified diagonal family; no images | 99.37 / 92.50 | 9.80 / 6.70 |
| LN-bias cleanup | Known location; published projection | 99.37 / 92.50 | 99.37 / 92.50 |
| Student + TA | 2048 target-domain queries; student training | 95.53 / 95.95 | 95.53 / 95.95 |
| Student + AGA target | 2048 target-domain queries; student training | 95.35 / 95.94 | 95.35 / 95.94 |
| Student + AGA both | 2048 target-domain queries; student training | 95.25 / 95.29 | 95.25 / 95.29 |

All merges use TA0.5. Diagonal adaptation is the official MLP kernel plus an explicitly specified native Attention adaptation, not a complete official attack release. Cleanup is a known-location diagnostic. Native FP32 cleanup changes some scores beyond the historical numerical-control tolerance; no exact FP32 function identity is claimed. Student rows share one training problem because all2048 queried hard labels coincide; they are not independent seed replications or the complete PaRaMS KD implementation.

Table 5: Supplementary mergers after original AGA on the protected expert

| Merger | Unprotected M / S | Defense 1 M / S | O/B [H] M / S | A/B [H] M / S |
|---|---|---|---|---|
| TA | 99.30 / 92.78 | 9.80 / 6.70 | 79.53 / 65.82 | 85.42 / 75.50 |
| TIES | 99.32 / 94.81 | 9.82 / 9.69 | 57.13 / 50.75 | 65.76 / 66.21 |
| DARE+TA | 99.30 / 92.81 | 24.20 / 25.50 | 79.52 / 66.01 | 86.11 / 75.87 |
| NUFILT M→S | 99.39 / 94.68 | 99.39 / 94.68 | 99.24 / 93.80 | 99.14 / 94.61 |
| NUFILT S→M | 99.07 / 95.35 | 99.07 / 95.35 | 98.97 / 95.07 | 98.86 / 95.38 |
| ProDistill | 99.42 / 96.83 | 9.80 / 6.70 | 99.42 / 96.16 | 99.42 / 96.46 |

MNIST target / SVHN partner. Each merger has its own matching control. DARE+TA is the mean of fixed mask seeds42/43/44; other entries are fixed runs. NUFILT reports both task orders and uses native adapters of official kernels. ProDistill uses64+64 unlabeled training calibration images and attacker-side elementwise coefficient training; it is not a data-free attack. Mergers are not counted as separate recovery attacks. [H] Task-bound failure retained.

Table 6: DARE mask variability under AGA target

| Protection | MNIST mean ± SD | SVHN mean ± SD |
|---|---|---|
| Unprotected | 99.30 ± 0.00 | 92.81 ± 0.04 |
| Defense 1 | 24.20 ± 5.60 | 25.50 ± 10.30 |
| O/B [H] | 79.52 ± 0.57 | 66.01 ± 0.71 |
| A/B [H] | 86.11 ± 0.95 | 75.87 ± 0.56 |

Sample standard deviation across the same three fixed masks42/43/44. This is mask variability, not three independent expert or defense trainings. All three mask values are retained in the cited source rows.

Table 7: Supplementary mergers: DTD and SVHN protected experts

| Target / partner | Merger / order | AGA scope | Control T / P | Defense 1 T / P |
|---|---|---|---|---|
| DTD / MNIST | NUFILT DTD / MNIST | None | 72.18 / 99.45 | 72.18 / 99.45 |
| DTD / MNIST | NUFILT DTD / MNIST | Target | 70.59 / 99.45 | 70.59 / 99.45 |
| DTD / MNIST | NUFILT DTD / MNIST | Both | 70.90 / 99.42 | 70.90 / 99.42 |
| DTD / MNIST | NUFILT MNIST / DTD | None | 72.71 / 99.51 | 72.71 / 99.51 |
| DTD / MNIST | NUFILT MNIST / DTD | Target | 70.96 / 99.51 | 70.96 / 99.51 |
| DTD / MNIST | NUFILT MNIST / DTD | Both | 71.44 / 99.48 | 71.44 / 99.48 |
| DTD / MNIST | ProDistill | None | 73.14 / 99.51 | 2.13 / 98.06 |
| DTD / MNIST | ProDistill | Target | 73.09 / 99.52 | 2.13 / 98.07 |
| DTD / MNIST | ProDistill | Both | 73.40 / 99.47 | 2.13 / 97.71 |
| SVHN / MNIST | NUFILT SVHN / MNIST | None | 95.11 / 99.19 | 95.11 / 99.19 |
| SVHN / MNIST | NUFILT SVHN / MNIST | Target | 94.40 / 99.25 | 94.40 / 99.25 |
| SVHN / MNIST | NUFILT SVHN / MNIST | Both | 94.51 / 99.14 | 94.51 / 99.14 |
| SVHN / MNIST | NUFILT MNIST / SVHN | None | 94.45 / 99.43 | 94.45 / 99.43 |
| SVHN / MNIST | NUFILT MNIST / SVHN | Target | 93.47 / 99.44 | 93.47 / 99.44 |
| SVHN / MNIST | NUFILT MNIST / SVHN | Both | 93.75 / 99.39 | 93.75 / 99.39 |
| SVHN / MNIST | ProDistill | None | 96.73 / 99.45 | 6.70 / 9.80 |
| SVHN / MNIST | ProDistill | Target | 96.45 / 99.45 | 6.70 / 9.80 |
| SVHN / MNIST | ProDistill | Both | 96.56 / 99.44 | 6.70 / 9.80 |

NUFILT uses both task orders. SVHN no-attack/both-AGA cells reuse the identical earlier MNIST/SVHN merger models with dataset roles exchanged; no new fits. The selected-parameter identity for Defense1 is retained. ProDistill uses64+64 unlabeled calibration images and official coefficient kernels. These are merger evaluations, not additional recovery attacks.

Table 8: Eight participating experts: three evaluated tasks

| Protected task | Protection | No recovery M / D / S | AGA target M / D / S |
|---|---|---|---|
| MNIST | Unprotected | 94.00 / 52.18 / 74.42 | 93.23 / 52.34 / 73.90 |
| MNIST | Defense 1 | 9.80 / 5.85 / 6.70 | 9.80 / 5.90 / 6.70 |
| MNIST | O/B [H] | 51.73 / 50.64 / 40.57 | 54.22 / 50.80 / 44.63 |
| MNIST | A/B [H] | 61.94 / 50.21 / 53.09 | 59.16 / 50.59 / 51.97 |
| MNIST | PaRaMS (full) | 9.82 / 2.13 / 9.65 | 93.23 / 52.34 / 73.90 |
| MNIST | PaRaMS attention ablation | 94.00 / 52.13 / 74.45 | 93.23 / 52.34 / 73.90 |
| DTD | Unprotected | 94.00 / 52.18 / 74.42 | 94.06 / 51.38 / 74.46 |
| DTD | Defense 1 | 72.22 / 2.13 / 52.48 | 72.30 / 2.13 / 52.47 |
| SVHN | Unprotected | 94.00 / 52.18 / 74.42 | 93.78 / 52.18 / 72.53 |
| SVHN | Defense 1 | 9.80 / 27.07 / 6.70 | 9.80 / 27.07 / 6.70 |

TA0.2 with eight experts; cells always MNIST / DTD / SVHN, including partner tasks. Other five task datasets were not evaluated. No eight-task average. [H] Original task-bound failures retained. Complete PaRaMS and its Attention ablation remain distinct.

Table 9: Paper-formula Unmask: a separate eight-expert recovery setting

| Protected task | Protection | No recovery M / D / S | Unmask0.3 M / D / S |
|---|---|---|---|
| MNIST | Unprotected | 94.00 / 52.18 / 74.42 | 7.52 / 1.97 / 9.55 |
| MNIST | Defense 1 | 9.80 / 5.85 / 6.70 | 10.28 / 2.07 / 7.76 |
| MNIST | O/B [H] | 51.73 / 50.64 / 40.57 | 11.35 / 1.97 / 17.59 |
| MNIST | A/B [H] | 61.94 / 50.21 / 53.09 | 10.89 / 2.13 / 12.85 |
| MNIST | PaRaMS (full) | 9.82 / 2.13 / 9.65 | 9.58 / 1.70 / 14.45 |
| MNIST | PaRaMS attention ablation | 94.00 / 52.13 / 74.45 | 7.51 / 1.97 / 9.55 |
| DTD | Unprotected | 94.00 / 52.18 / 74.42 | 10.00 / 2.55 / 9.56 |
| DTD | Defense 1 | 72.22 / 2.13 / 52.48 | 10.28 / 2.13 / 7.76 |
| SVHN | Unprotected | 94.00 / 52.18 / 74.42 | 7.91 / 2.13 / 9.15 |
| SVHN | Defense 1 | 9.80 / 27.07 / 6.70 | 11.35 / 2.23 / 19.59 |

TA0.2 with eight experts; cells always MNIST / DTD / SVHN, including partner tasks. Other five task datasets were not evaluated. No eight-task average. [H] Original task-bound failures retained. Complete PaRaMS and its Attention ablation remain distinct. Fixed merged parameters minus0.3 times the whole published expert. Both protected and unprotected controls receive the operation. Paper-formula native adaptation, not official code or a complete paper table. No AGA composition. The unprotected model also collapses; low absolute scores do not demonstrate successful expert recovery.

Table 10: Targeted bias cleanup: numerical protocols kept separate

| Target / partner | Numerics | Scope | Control T / P | Defense 1 T / P |
|---|---|---|---|---|
| MNIST / SVHN | Native FP32 | Target | 99.37 / 92.50 | 99.37 / 92.50 |
| MNIST / SVHN | Native FP32 | Both | 99.37 / 92.50 | 99.37 / 92.50 |
| MNIST / DTD | Native FP32 | Target | 99.45 / 69.26 | 99.45 / 69.26 |
| MNIST / DTD | Native FP32 | Both | 99.45 / 69.26 | 99.45 / 69.26 |
| DTD / MNIST | FP32 prefix; FP64 readout | Target | 69.26 / 99.45 | 69.26 / 99.45 |
| DTD / MNIST | FP32 prefix; FP64 readout | Both | 69.26 / 99.45 | 69.26 / 99.45 |
| SVHN / MNIST | FP32 prefix; FP64 readout | Target | 92.50 / 99.37 | 92.50 / 99.37 |
| SVHN / MNIST | FP32 prefix; FP64 readout | Both | 92.50 / 99.37 | 92.50 / 99.37 |

TA0.5; no images, original expert or secret perturbation enters recovery construction. The FP64-readout rows are a separately frozen protocol and must not be pooled with native FP32. Original native DTD control failure0.00082493 and MNIST before/after score difference0.00069427 remain recorded. Equal recovered predictions do not establish exact FP32 function identity.

Table 11: Task-direction changes and actual classification effects

| Fixed model | Task-inside distance | Task-outside distance | Standalone M | TA M | AGA both M |
|---|---|---|---|---|---|
| Global permutation [K] | 2.69895 | 11.86048 | 99.69 | 7.39 | 10.09 |
| Latest compensation [F] | 0.49964 | 11.85437 | 99.69 | 99.13 | 99.04 |

Same74 matrices and original expert-minus-base rank32 right singular directions. Squared per-matrix relative norms are summed. These matrix-only distances exclude biases/LN/embeddings and are not the full-model distance. Comparable task-outside distances do not imply comparable protection. [K] Known symmetry; [F] failed efficacy, no novelty claim.

Table 12: Latest Defense 2 candidate: source-only ablations

| Counterfactual | Correct | Prediction changes | Max. relative gap change |
|---|---|---|---|
| Task-inside matrices only | 511/512 | 1 | 1.05992 |
| Task-outside matrices only | 503/512 | 9 | 1.82071 |
| All changed matrices | 505/512 | 7 | 1.87001 |
| Other parameters only | 505/512 | 7 | 1.85216 |
| Complete joint change | 512/512 | 0 | 3.02985e-06 |

The fixed512 target-training source images, not a test table. Matrix inside/outside components use the same original rank32 task basis; non-matrix changes are separate. Each counterfactual is quantized separately. Complete-release replay and independent NumPy recount both have zero error. No candidate was repaired or selected from these counterfactual outcomes.

Table 13: Completed work and remaining comparisons

| Item | Status |
|---|---|
| TA + original AGA | Completed: two-expert pairs; target/both recovery scopes; protected and partner accuracy. |
| Defense 1 | MNIST, DTD, SVHN completed; targeted cleanup and fixed-budget hard-label recovery completed with separate permissions/numerics. |
| Defense 2 | Effective fixed data-free structural scheme verified on MNIST/DTD/SVHN. Task-SVD-guided construction and new method contribution remain under study. Later failed candidates and historical O/B/A/B task-bound failures remain distinct. |
| PaRaMS | Full-default protection completed for MNIST/DTD/SVHN, including matched TA and original AGA. Attention-only ablation completed for MNIST; diagonal recovery is a native adaptation. |
| MergeLock | Not run: official repository currently contains only README/Coming Soon. No invented implementation or paper scores. |
| MergeGuard / ARF | Full defenses not run; training-stage conditions differ. Unmask formula was tested separately; GradErase not reproduced. |
| MergeBarrier | Full defense not run; architecture adaptation and verified implementation missing. |
| Eight-task benchmark | Eight experts participated; protected/evaluated targets cover MNIST/DTD/SVHN. Complete protected cases and evaluation for SUN397/Cars/RESISC45/EuroSAT/GTSRB are missing. |
| Other mergers | TIES, three DARE masks, NUFILT and ProDistill completed in documented scopes. Full TSV-Merge, CAT and LOT not run. |
| Reliability | Per-class/pair counts and source audits retained. Only DARE has the stated three-mask comparison; do not present deterministic rows as multi-seed results. |

This is a completion inventory, not an accuracy table. Missing methods are not assigned zero accuracy. More merger names do not increase the number of recovery attacks.

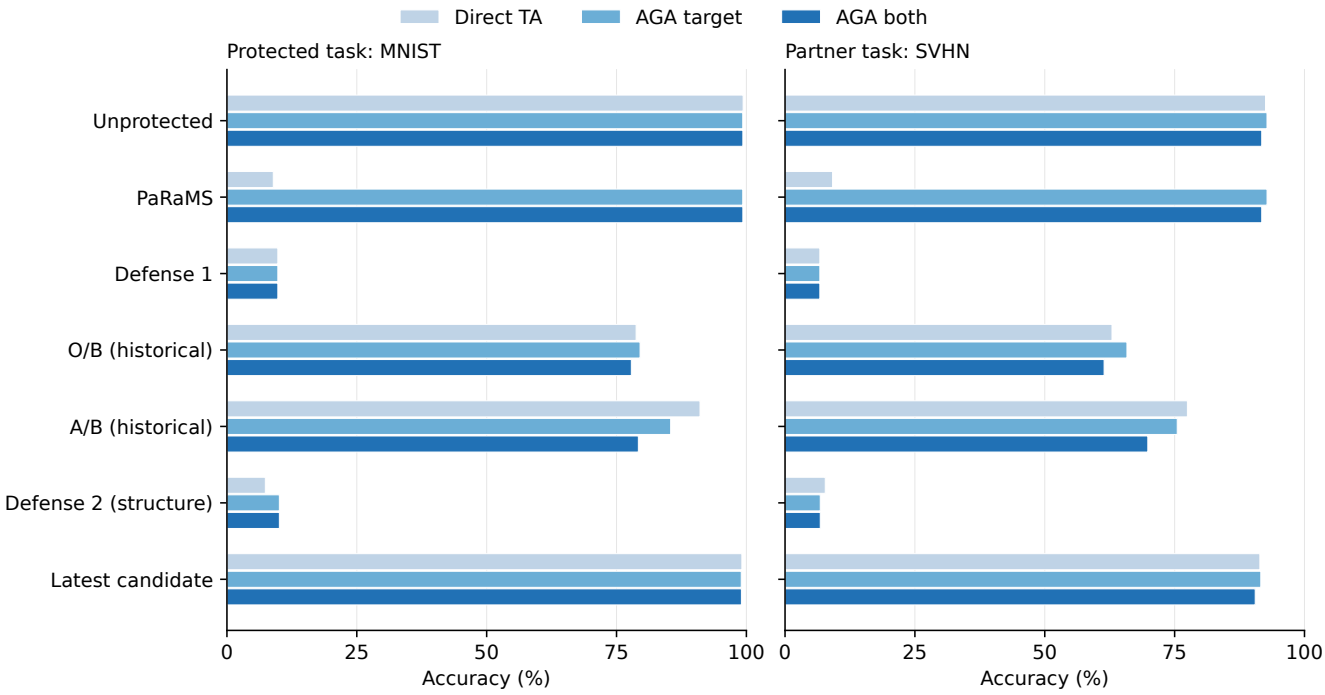


Figure 1: Two-expert TA0.5 before and after original AGA. Both protected and partner tasks are shown. Every bar is one fixed configuration; historical O/B and A/B task-bound failures remain labelled. Exact numbers appear in Table 1.